

1. Cars A and B move in the same direction in adjacent lanes. The position x of car A is given in the figure, from time $0 \leq t \leq 7.0\text{s}$. In the figure, $x_s = 32.0\text{ m}$. At $t = 0$, car B is at $x = 0$, with a velocity of 12 m/s and a negative constant acceleration a_B . What must a_B be such that the cars are (momentarily) side by side (at the same value of x) at $t = 4.0\text{ s}$?

Karre A en B beweeg in dieselfde rigting. Die posisie x van kar A word gegee in die figuur vir tyd $0 \leq t \leq 7\text{s}$.

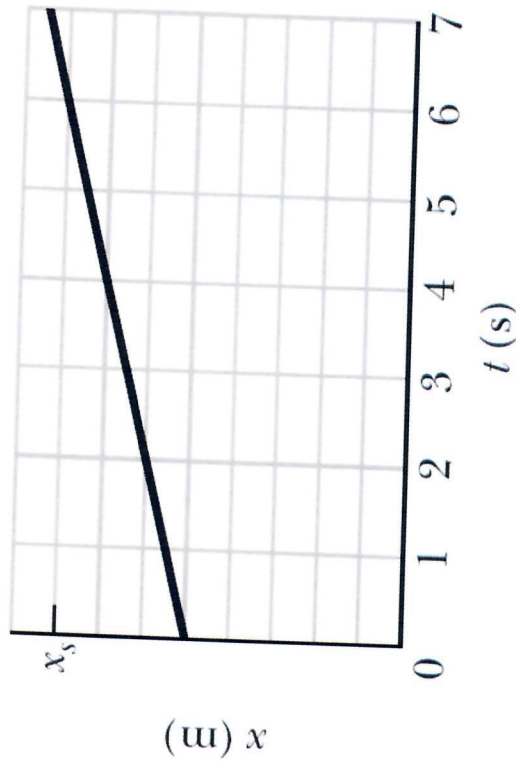
In die figuur is $x_s = 32.0\text{ m}$. By $t = 0$, is kar B by $x = 0$, en het 'n snelheid van 12 m/s en 'n negatiewe konstante versnelling a_B .

Wat moet a_B wees sodat die karre vir 'n oomblik lang mekaar is ($d/s\ x$ waarde) teen $t = 4.0\text{ s}$?

2. Two vectors are given by $\vec{A} = (4.0\text{m})\hat{i} - (3.0\text{m})\hat{j} + (1.0\text{m})\hat{k}$ and $\vec{B} = (-1.0\text{m})\hat{i} + (1.0\text{m})\hat{j} + (4.0\text{m})\hat{k}$. In unit vector notation, find the third vector $\vec{C}$, such that $\vec{A} - \vec{B} + \vec{C} = 0$. (5)

Two vectors are given by $\vec{A} = (4.0\text{m})\hat{i} - (3.0\text{m})\hat{j} + (1.0\text{m})\hat{k}$ and $\vec{B} = (-1.0\text{m})\hat{i} + (1.0\text{m})\hat{j} + (4.0\text{m})\hat{k}$. In unit vector notation, find the third vector $\vec{C}$, such that $\vec{A} - \vec{B} + \vec{C} = 0$. (5)

Twee vektore word gegee deur $\vec{A} = (4.0\text{m})\hat{i} - (3.0\text{m})\hat{j} + (1.0\text{m})\hat{k}$ en $\vec{B} = (-1.0\text{m})\hat{i} + (1.0\text{m})\hat{j} + (4.0\text{m})\hat{k}$. In eenheidsvektor notasie, vind die derde vektor $\vec{C}$, sodat $\vec{A} - \vec{B} + \vec{C} = 0$. (5)



Tutorial Test 2-1

1. From the graph: Velocity of A is constant.

At $t = 4s$, car A is at 28m. ✓

This is where car B must be at $t = 4s$.

Car B: $x_{iB} = 0$, $v_{iB} = 12 \text{ m/s}$
 $t = 4s$, $x_{fB} = 28 \text{ m}$ ✓

Using

$$x_{fB} = x_{iB} + v_{iB}t + \frac{1}{2}a_B t^2$$

$$\begin{aligned} a_B &= \frac{[x_{fB} - x_{iB}] - v_{iB}t}{t^2} \quad \checkmark \\ &= \frac{[(28\text{m} - 0\text{m}) - (12\text{m/s})(4\text{s})]}{(4\text{s})^2} \\ &= \textcircled{-2.5} \text{ m/s}^2 \quad \checkmark \end{aligned}$$

Tutors:

- Students must first show original formula
- Units! Units! Units! At all times.

Please check my calculations - Work the problems out for yourself

$$\begin{aligned}
 2. \quad \vec{A} &= 4.0\text{m} \hat{i} - 3.0\text{m} \hat{j} + 1.0\text{m} \hat{k} \\
 \vec{B} &= -1.0\text{m} \hat{i} + 1.0\text{m} \hat{j} + 4.0\text{m} \hat{k} \\
 \vec{C} &= x \hat{i} + y \hat{j} + z \hat{k}
 \end{aligned}$$

$$A_x - B_x + C_x = 0$$

$$C_x = -A_x + B_x = -4.0\text{m} + (-1.0\text{m})$$

$$C_x = -5.0\text{m} \checkmark$$

$$A_y - B_y + C_y = 0$$

$$C_y = -A_y + B_y = -(-3.0\text{m}) + 1.0\text{m}$$

$$C_y = 4.0\text{m} \checkmark$$

$$A_z - B_z + C_z = 0$$

$$C_z = -A_z + B_z = -(1.0\text{m}) + 4.0\text{m}$$

$$C_z = 3.0\text{m} \checkmark$$

$$\vec{C} = -5.0\text{m} \underline{\underline{\hat{i}}} + 4.0\text{m} \underline{\underline{\hat{j}}} + 3.0\text{m} \underline{\underline{\hat{k}}} \checkmark \checkmark$$

- 1(a) If the maximum acceleration that is tolerable for passengers in a subway train is 1.34 m/s^2 and subway stations are located 806 m apart, what is the maximum speed a subway train can attain between stations?

Indien die maksimum versnelling wat die passasiers in 'n moltrein gemaklik kan verdra is 1.34 m/s^2 is en die moltrein stasis 806 m uitmekaar is, wat is die maksimum speed wat die moltrein kan bereik tussen stasies?

- (b) What is the travel time between stations? *Wat is die reistyd tussen stasies?* (5)

2. Two vectors are given by $\vec{A} = (4.0\text{m})\hat{i} - (3.0\text{m})\hat{j} + (1.0\text{m})\hat{k}$ and $\vec{B} = (-1.0\text{m})\hat{i} + (1.0\text{m})\hat{j} + (4.0\text{m})\hat{k}$. In unit vector notation, find the third vector $\vec{C}$, such that $\vec{A} + \vec{B} + \vec{C} = 0$.

Twee vektore word gegee deur $\vec{A} = (4.0\text{m})\hat{i} - (3.0\text{m})\hat{j} + (1.0\text{m})\hat{k}$ en

$\vec{B} = (-1.0\text{m})\hat{i} + (1.0\text{m})\hat{j} + (4.0\text{m})\hat{k}$. In eenheidsvektor notasie, vind die derde vektor $\vec{C}$, sodat $\vec{A} + \vec{B} + \vec{C} = 0$. (5)

Tutorial Test 2-2

1. $a_{\max} = 1.34 \text{ m/s}^2$

$$\Delta x_{\text{total}} = 806 \text{ m}$$

$$v_{\max} = ?$$

$$v_i = 0 \text{ m/s} \checkmark$$

$$t_i = 0 \text{ s}$$

$$t_f = ?$$

a) Distance to max. speed: $x_f = \frac{\Delta x_{\text{total}}}{2}$
 $= 403 \text{ m} \checkmark$

Using $v_{xf}^2 = v_{xi}^2 + 2a_x \Delta x$
 $= 0 + 2(1.34 \text{ m/s}^2)(403 \text{ m} - 0 \text{ m})$

$$v_{\max} = v_{xf} = 32.8 \text{ m/s} \checkmark$$

b) $x_f = x_i + v_{ix} t + \frac{1}{2} a_x t^2$

$$\frac{2 \Delta x}{a_x} = t^2 \checkmark = \frac{2(403 \text{ m})}{1.34 \text{ m/s}^2}$$

$$t = 24.5 \text{ s} \checkmark \text{ (time to max speed)}$$

Total time between stations

$$t_{\text{total}} = 2(24.5) = 49 \text{ s} \checkmark$$

$$\begin{aligned}
 2. \quad \vec{A} &= 4.0\text{m} \hat{i} - 3.0\text{m} \hat{j} + 1.0\text{m} \hat{k} \\
 \vec{B} &= -1.0\text{m} \hat{i} + 1.0\text{m} \hat{j} + 4.0\text{m} \hat{k} \\
 \vec{C} &= x \hat{i} + y \hat{j} + z \hat{k}
 \end{aligned}$$

$$A_x + B_x + C_x = 0$$

$$C_x = -A_x - B_x = -4.0\text{m} - (-1.0\text{m})$$

$$C_x = -3.0\text{m} \checkmark$$

$$A_y + B_y + C_y = 0$$

$$C_y = -A_y - B_y = -(-3.0\text{m}) - (1.0\text{m})$$

$$C_y = 3.0\text{m} \checkmark$$

$$A_z + B_z + C_z = 0$$

$$C_z = -A_z - B_z = -1.0\text{m} - 4.0\text{m}$$

$$= -5.0\text{m} \checkmark$$

$$\vec{C} = -3.0\text{m} \hat{i} + 3.0\text{m} \hat{j} - 5.0\text{m} \hat{k} \checkmark \checkmark$$

1. A key falls from a bridge that is 45 m above the water. It falls directly into a model boat, moving with constant velocity, is 12 m from the point of impact when the key is released. What is the speed of the boat?

'n Sleutel val van 'n brug af 45 m bo die water. Dit val in 'n model bootjie wat teen 'n konstante snelheid beweeg. Die bootjie is 12 m van die impak-punt wanneer die sleutel gelos word. Hoe vinnig beweeg die bootjie?

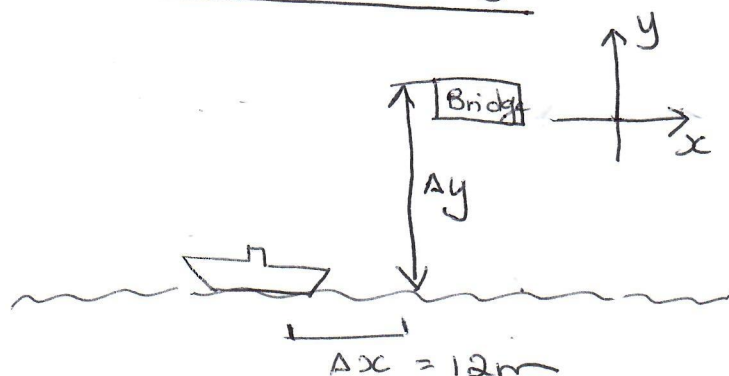
(5)

2. If $\vec{B}$ is added to $\vec{C} = (3.0)\hat{i} + (4.0\text{m})\hat{j}$, the result is a vector in the positive direction of the y -axis, with a magnitude equal to that of $\vec{C}$. What is the magnitude of $\vec{B}$?

Indien $\vec{B}$ by vector $\vec{C} = (3.0)\hat{i} + (4.0\text{m})\hat{j}$ by getel word is die resultante vector in die rigting van die positiewe y -as, en dit het 'n grootte wat gelyk is aan vector $\vec{C}$. Wat is die grootte van $\vec{B}$?

(5)

Tutorial Test 2-3



$$a_y = -9.8 \text{ m/s}^2 \quad \checkmark$$

$$v_{y0} = 0 \text{ m/s}$$

$$\Delta y = -45 \text{ m}$$

$$t_{\text{fall}} = t_{\text{boat}}$$

$$a_x = 0 \text{ m/s}$$

$$v_x = \text{constant} = ?$$

$$\Delta x = 12 \text{ m}$$

$$\Delta x = v_{x0} t$$

NB: If the student uses

$a = +9.8 \text{ m/s}$ and
has not indicated
↓ (y +ve downward)

it must be marked
incorrect.

$$\Delta y = v_{y0} t + \frac{1}{2} a_y t^2$$

Solve for t using the
formula

OR

$$\begin{aligned} v_{fy}^2 &= v_{y0}^2 + 2a_y \Delta y \\ &= 0 + 2(-9.8 \text{ m/s}^2)(-45 \text{ m}) \\ &= -29.7 \text{ m/s} \quad \checkmark \end{aligned}$$

$$\begin{aligned} v_{fy} &= v_{y0} + a t \\ t &= \frac{v_{fy} - v_{y0}}{a} \\ &= \frac{-29.7 \text{ m/s} - 0}{-9.8 \text{ m/s}^2} \\ &= +3.03 \text{ s} \quad \checkmark \end{aligned}$$

Using $\Delta x = v_{x0} t$

$$\begin{aligned} v_{x0} &= \frac{\Delta x}{t} \\ &= 12 \text{ m} / 3.03 \text{ s} \end{aligned}$$

$$v_{x0} = 3.96 \text{ m/s} \quad \checkmark \quad \text{for the boat}$$

2

2-3.

$$\vec{C} + \vec{B} = \vec{R}$$

$$\vec{C} = 3.0\hat{i} + 4.0\hat{j}$$

$$\vec{B} = x_B\hat{i} + y_B\hat{j}$$

$$\vec{R} = 0\hat{i} + 5.0\hat{j} \quad \checkmark$$

$$\text{Magnitude of } |\vec{C}| = \sqrt{C_x^2 + C_y^2} = \sqrt{3^2 + 4^2} \\ = 5.0 \quad \checkmark$$

$$R_x = 3.0 + x_B = 0$$

$$\therefore x_B = -3.0 \quad \checkmark$$

$$R_y = 4.0 + y_B = 5.0$$

$$\therefore y_B = 1.0 \quad \checkmark$$

$$\vec{B} = -3.0\hat{i} + 1.0\hat{j}$$

$$|\vec{B}| = \sqrt{B_x^2 + B_y^2} = \sqrt{(-3.0)^2 + (1.0)^2}$$

$$= \sqrt{10} = 3.16$$

$$|B| = 3.2 \quad \checkmark$$

1. At the instant the traffic light turns green, an automobile starts from rest with a constant acceleration of 2.2 m/s^2 . At the same instant a truck, traveling with a constant speed of 9.5 m/s , overtakes and passes the automobile. (a) How far beyond the traffic signal will the automobile overtake the truck? (b) How fast will the automobile be traveling at that instant?

Die oomblik dat die verkeerslig na groen oorslaan begin 'n kar beweeg teen 'n konstante versnelling van 2.2 m/s^2 . Op dieselfde oomblik ry 'n trok verby die kar met 'n konstante spoed van 9.5 m/s . (a) Hoe ver verby die verkeerslig sal die kar die trok verbystee? (b) Hoe vinnig sal die kar beweeg op daardie oomblik?

(5)

2. If $\vec{B}$ is added to $\vec{C} = (3.0)\hat{i} + (4.0\text{m})\hat{j}$, the result is a vector in the positive direction of the y -axis, with a magnitude equal to that of $\vec{C}$. What is the magnitude of $\vec{B}$?

Indien $\vec{B}$ by vector $\vec{C} = (3.0)\hat{i} + (4.0\text{m})\hat{j}$ by getel word is die resultante vector in die rigting van die positiewe y -as, en dit het 'n grootte wat gelyk is aan vector $\vec{C}$. Wat is die grootte van $\vec{B}$?

(5)

Tutorial Test 2-4

$$1. \quad a_{\text{car}} = 2.2 \text{ m/s}^2$$

$$v_{\text{car},i} = 0 \text{ m/s}$$

$$a_{\text{truck}} = 0 \text{ m/s}^2$$

$$v_i = 9.5 \text{ m/s}$$

When car passes truck

$$\Delta x_{\text{car}} = \Delta x_{\text{truck}}$$

$$t_{\text{car}} = t_{\text{truck}}$$

a) Displacement for both car and truck:

$$\Delta x = \underbrace{v_{\text{car},i} t}_{=0} + \frac{1}{2} a_{\text{car}} t^2 = \underbrace{v_{\text{truck},i} t}_{=0} + \frac{1}{2} a_{\text{truck}} t^2 \quad \checkmark$$

$$\therefore t = \frac{2v_{\text{truck},i}}{a_{\text{car}}} \quad \checkmark$$

$$= \frac{2(9.5 \text{ m/s})}{2.2 \text{ m/s}^2}$$

$$= 8.6 \text{ s} \quad \checkmark$$

$$\text{Therefore } \Delta x = v_{\text{truck},i} t = (9.5 \text{ m/s})(8.6 \text{ s})$$

$$\Delta x = 82 \text{ m} \quad \checkmark$$

$$b) \quad v_{\text{car},f} = v_{\text{car},i} + a_{\text{car}} t$$

$$= 0 + (2.2 \text{ m/s}^2)(8.6 \text{ s})$$

$$= 18.9 \text{ m/s} = 19 \text{ m/s} \quad \checkmark$$

2

$$\vec{C} + \vec{B} = \vec{R}$$

$$\vec{C} = 3.0\hat{i} + 4.0\hat{j}$$

$$\vec{B} = x_B\hat{i} + y_B\hat{j}$$

$$\vec{R} = 0\hat{i} + r_y\hat{j} \quad \checkmark$$

Magnitude of $|\vec{C}| = \sqrt{C_x^2 + C_y^2} = \sqrt{3^2 + 4^2}$
 $= 5.0 \quad \checkmark$

$$R_x = 3.0 + x_B = 0$$

$$\therefore x_B = -3.0 \quad \checkmark$$

$$R_y = 4.0 + y_B = 5.0$$

$$\therefore y_B = 1.0 \quad \checkmark$$

$$\vec{B} = -3.0\hat{i} + 1.0\hat{j}$$

$$|\vec{B}| = \sqrt{B_x^2 + B_y^2} = \sqrt{(-3.0)^2 + (1.0)^2}$$

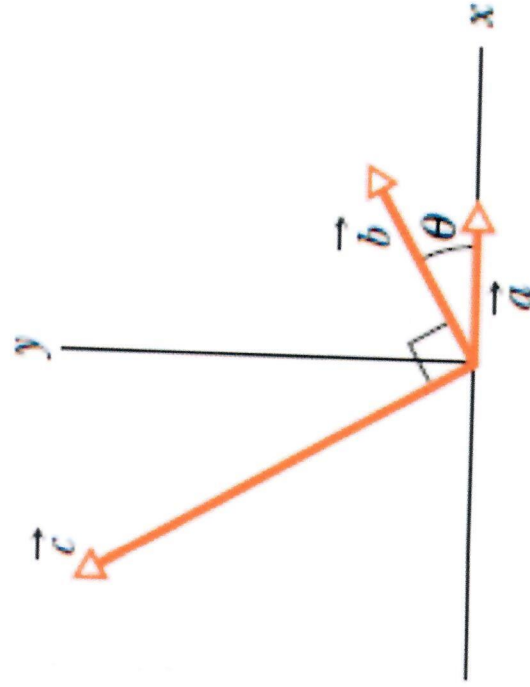
$$= \sqrt{10} = 3.16$$

$$|B| = 3.2 \quad \checkmark$$

1. A rock is thrown vertically upward from ground level at time $t = 0$. At $t = 1.5$ s it passes the top of a tall tower, and 1.0 s later it reaches its maximum height. What is the height of the tower? 'n Klip word vertikaal opwaarts gegooi vanaf die grond by $t = 0$. Teen $t = 1.5$ s gaan die klip verby die bokant van 'n toring, en 1.0 s later bereik dit 'n maksimum hoogte. Wat is die hoogte van die toring? (5)

2. The three vectors in the figure have magnitudes $a = 3.00$ m, $b = 4.00$ m, and $c = 10.0$ m and the angle $\theta = 30.0^\circ$. Calculate the components of the vectors and determine the sum of the three vectors.

Die drie vektore in die figuur het groottes $a = 3.00$ m, $b = 4.00$ m, en $c = 10.0$ m en die hoek $\theta = 30.0^\circ$. Bereken die komponente van die vektore en bereken dan die som van die drie vektore. (5)



Tutorial 2-5

$$\begin{aligned}\text{At } t = 0 \quad y &= 0 \\ v_{y0} &= ? \\ a &= -9.8 \text{ m/s}^2\end{aligned}$$

$$\begin{aligned}\text{At } t = 2.5 \text{ s} \quad y &= y_{\text{max}} \\ v_y &= 0 \text{ m/s} \\ a &= -9.8 \text{ m/s}^2\end{aligned}$$

We need to find the initial velocity
At $t = 2.5 \text{ s}$

$$v_{fy} = v_{iy} + a_y t$$

$$v_{iy} = + v_{fy} - a_y t \quad \checkmark$$

$$= \textcircled{0} \checkmark - (-9.8 \text{ m/s}^2)(2.5 \text{ s}) \checkmark$$

$$= 24.5 \text{ m/s} \quad \checkmark$$

Height of the tower at $t = 1.5 \text{ s}$

$$\Delta y = v_{iy} t + \frac{1}{2} a_y t^2$$

$$= 24.5 \text{ m/s} + \frac{1}{2} (-9.8 \text{ m/s}^2)(1.5 \text{ s})^2$$

$$= 25.7 \text{ m}$$

$$= 26 \text{ m} \quad \checkmark$$

$$2. \quad a_x = 3.00 \text{ m} \hat{i} \checkmark$$

$$b_x = b \cos 30^\circ \\ = 3.46 \text{ m} \checkmark$$

$$c_x = c \cos 120^\circ \\ = -5.00 \text{ m} \checkmark$$

$$a_y = 0 \checkmark$$

$$b_y = b \sin 30^\circ \\ = 2.00 \text{ m} \checkmark$$

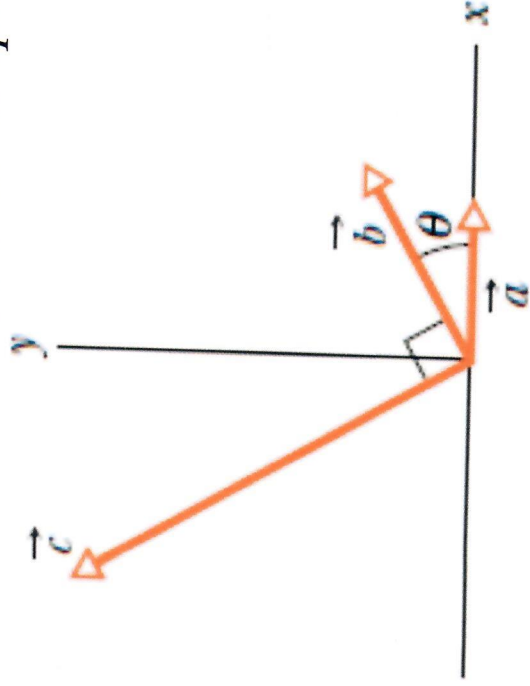
$$c_y = c \sin 120^\circ \\ = 8.66 \text{ m} \checkmark$$

$$\vec{a} + \vec{b} + \vec{c} = 1.46 \text{ m} \hat{i} + 10.66 \text{ m} \hat{j}$$

1. A rock is thrown vertically upward from ground level at time $t = 0$. At $t = 1.5$ s it passes the top of a tall tower, and 1.0 s later it reaches its maximum height. What is the height of the tower?
 'n Klip word vertikaal opwaarts gegooi vanaf die grond by $t = 0$. Teen $t = 1.5$ s gaan die klip verby die bokant van 'n toring, en 1.0 s later bereik dit 'n maksimum hoogte. Wat is die hoogte van die toring? (5)

2. The three vectors in the figure have magnitudes $a = 3.00$ m, $b = 4.00$ m, and $c = 10.0$ m and the angle $\theta = 30.0^\circ$. Calculate the components of the vectors and determine the sum of the three vectors.

Die drie vektore in die figuur het groottes $a = 3.00$ m, $b = 4.00$ m, en $c = 10.0$ m en die hoek $\theta = 30.0^\circ$. Bereken die komponente van die vektore en bereken dan $\vec{a} + 2\vec{b} - \vec{c}$. (5)



Tutorial 2-6

At $t = 0$

$$y = 0$$

$$v_{y0} = ?$$

$$a = -9.8 \text{ m/s}^2$$

At $t = 2.5 \text{ s}$

$$y = y_{\text{max}}$$

$$v_y = 0 \text{ m/s}$$

$$a = -9.8 \text{ m/s}^2$$

We need to find the initial velocity
At $t = 2.5 \text{ s}$

$$v_{fy} = v_{iy} + at$$

$$v_{iy} = +v_{fy} - at \quad \checkmark$$

$$= \textcircled{0} \checkmark - (-9.8 \text{ m/s}^2)(2.5 \text{ s}) \quad \checkmark$$

$$= 24.5 \text{ m/s} \quad \checkmark$$

Height of the tower at $t = 1.5 \text{ s}$

$$\Delta y = v_{iy}t + \frac{1}{2}a_yt^2$$

$$= 24.5 \text{ m/s} + \frac{1}{2}(-9.8 \text{ m/s}^2)(1.5 \text{ s})^2$$

$$= 25.7 \text{ m}$$

$$= 26 \text{ m} \quad \checkmark$$

$$2. \quad a_x = 3.00 \text{ m} \hat{i} \checkmark$$

$$a_y = 0 \checkmark$$

$$b_x = b \cos 30^\circ \\ = 3.46 \text{ m} \checkmark$$

$$b_y = b \sin 30^\circ \\ = 2.00 \text{ m} \checkmark$$

$$c_x = c \cos 120^\circ \\ = -5.00 \text{ m} \checkmark$$

$$c_y = c \sin 120^\circ \\ = 8.66 \text{ m} \checkmark$$

$$\vec{a} + 2\vec{b} - \vec{c} = 14.92 \text{ m} \hat{i} - 4.66 \text{ m} \hat{j}$$

My apologies for the error on the question. If you solved the English Version, the solution is the same as in Test 2-5.